Linear Regression & Classification

McGill Computer Science

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1 Abstract

This project evaluates the performance of linear regression (multiple and multivariate), binary logistic regression, and multi-class logistic regression on two benchmark classification tasks. For binary classification, we use the Breast Cancer Wisconsin Diagnostic dataset to predict tumor malignancy, and for multi-class classification, we use the Wine Recognition dataset to classify three different types of wine. We found that both linear and logistic regression performed very well, with our linear regression models outperforming our logistic regression models slightly for both datasets. Overall, our results demonstrate the importance of model choice, feature selection, and feature transformation to maximize classification accuracy.

2 Introduction

This study explores linear modeling techniques for classification by implementing and comparing multiple linear regression, multivariate linear regression, binary logistic regression, and multi-class logistic regression using two datasets: the Breast Cancer Wisconsin Diagnostic dataset and the Wine Recognition dataset. The binary task is based on the Breast Cancer dataset, where the goal is to distinguish between malignant and benign tumors. It has been used in studies exploring the use of machine learning techniques such as General Linear Model regression (GLMs) and single-layer Artificial Neural Networks to develop predictive models for cancer diagnosis [4]. The multi- class task uses the Wine Recognition dataset to classify wines into three categories based on continuous features such as chemical constituents. The Wine dataset has been used in recent studies to explore various analytical methods, such as clustering via the optimal transport barycenter problem [5] and testing an information criterion for selecting auxiliary variables in incomplete data analysis [2]. Our approach begins with data preprocessing and exploratory analysis of feature importance using simple regression, which aids in filtering features when training the models. We then implement closed-form solutions for linear regression and gradient-based approaches for the logistic models. By analyzing convergence behavior, gradient accuracy, and performance metrics, we provide insight into the trade-offs between these three models. We found that for the binary classification task, linear regression outperformed binary logistic regression, and for the multi-class classification, multiple and multivariate linear regression outperformed multi-class logistic regression. However, transforming features using a second-degree polynomial improved the accuracy of both of our logistic regression models.

3 Datasets

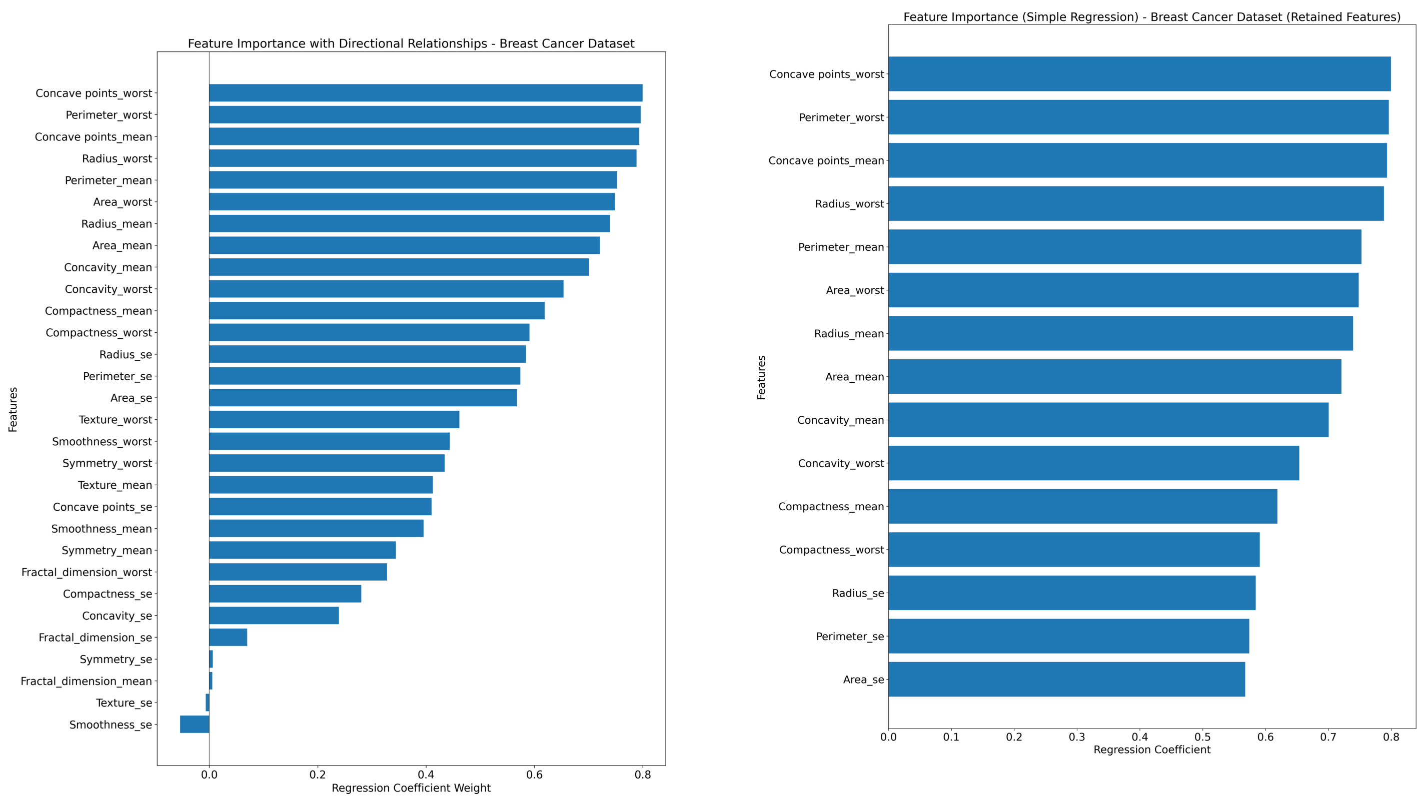
For our binary classification task, we used the Breast Cancer Wisconsin Diagnostic dataset, which contains 569 samples with 30 continuous features derived from digitized images of fine needle aspirate biopsies of breast masses. The dataset originally included an ID column and diagnosis labels (M for malignant and B for benign), so we removed the ID feature and converted the labels to binary values (1 for malignant and 0 for benign). All features were standardized to have zero mean and unit variance, and the dataset was split into training, validation, test sets using stratified sampling to ensure that each set maintains approximately the same proportion of classes as the original dataset. To assess feature relevance, we computed simple regression coefficients using the expression $w_d = \frac{x_d^\top y}{N}$, which after standardization acts equivalently to calculating covariance between each feature and the target. These coefficients were then ranked from most positive to most negative, revealing that features such as Concave points worst, Perimeter worst, and Concave points mean were the most influential (Fig. 1a). Using a mean threshold of the absolute coefficients, we filtered the dataset to only retain features with importance scores above this value, leaving us with 15 of the original 30 features for subsequent steps (Fig. 1b).

For our multi-class classification task, we used the Wine Recognition dataset, which contains data from a chemical analysis of wines grown in similar regions in Italy but produced from three different cultivars. There are 178 samples, and the analysis found 13 different chemical constituents in the three types of wines, which act as the features. The class labels for the types of wines were originally 1, 2, 3, so we converted them to a zero-based index (0, 1, 2) and the features were standardized similarly to the binary task. Likewise, the dataset was then divided into training, validation, and test sets via stratified sampling. We computed feature importance by one- hot encoding the target labels and applying the regression formula $W = \frac{X^{\top} Y}{N}$, resulting in a $D \times C$ coefficient matrix. By extracting the mean absolute coefficient for each feature and ranking them, we found that the most important features were Proline, Color intensity, OD280/OD315 of diluted wines, Alcohol, Flavanoids, Hue, and Total phenols (Fig. 2a). Directional feature importance for each individual class was also visualized (Fig. 2b). Only those features with importance values above the calculated mean threshold were retained, reducing the feature set from 13 to 7 (Fig. 2a). Filtering of the dataset simplifies the model and helps eliminate noise to prevent decreased accuracy and poor generalization.

In order to use our regression models for classification, we transformed their continuous outputs into probability estimates. For the binary classification task with the Breast Cancer dataset, we applied the sigmoid function, which maps any real-valued input into the interval [0,1]. This conversion allows us to interpret the output as a probability of the positive class and compute metrics such as AUROC. For the multiclass classification task with the Wine Recognition dataset, we used the softmax function to transform the raw outputs into a probability distribution over the three classes. These transformations ensure that the regression outputs are directly comparable to the probabilities produced by standard classification models such as KNN or DT.

4 Results

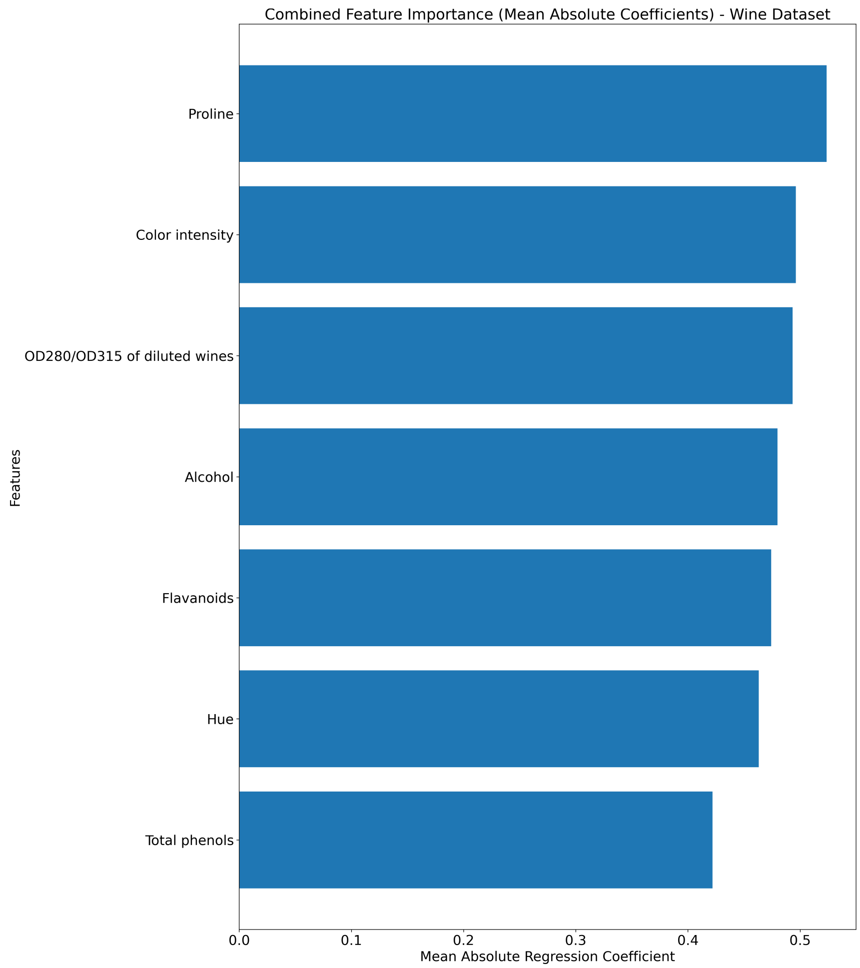
In our experiments, we first computed feature importance on both datasets using simple linear regression, outlined in the Datasets section. Features were plotted from most positive regression coefficient to most negative, which can be seen in Figures 1 and 2.



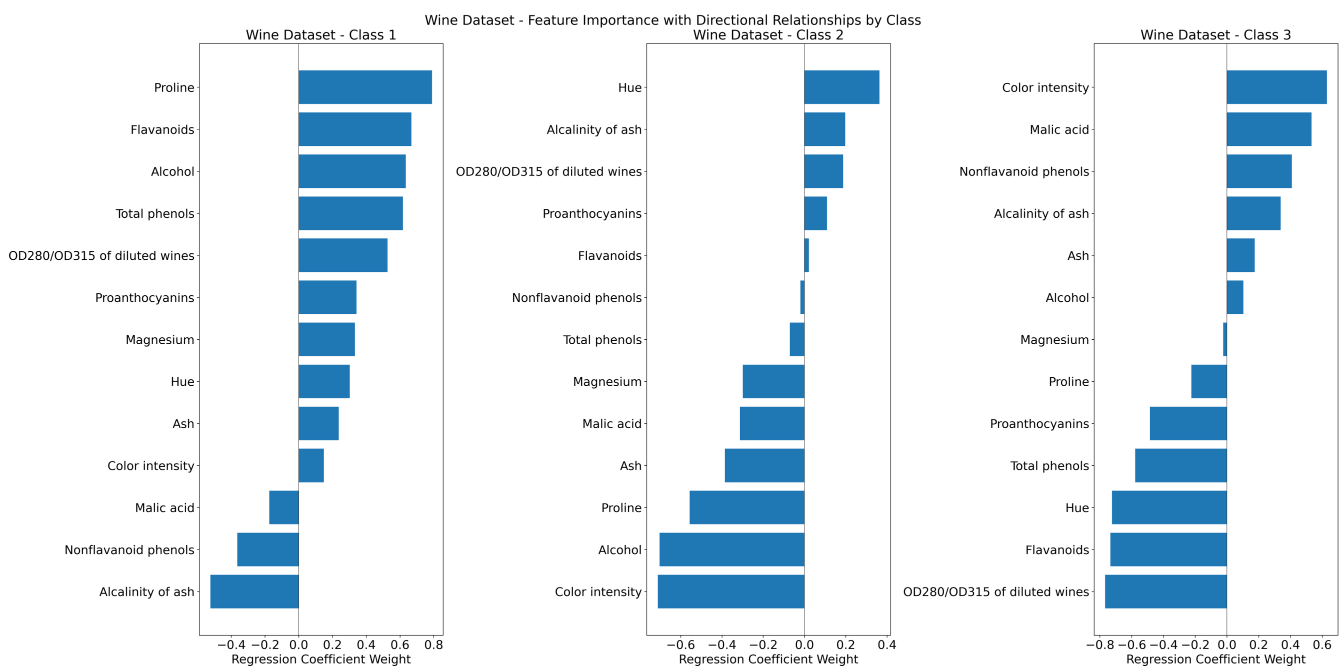
(a) Ranking of features for the Breast Cancer dataset.

(b) Retained features above mean threshold for the Breast Cancer Dataset.

Figure 1: Feature Importance Rankings for the Breast Cancer Dataset.



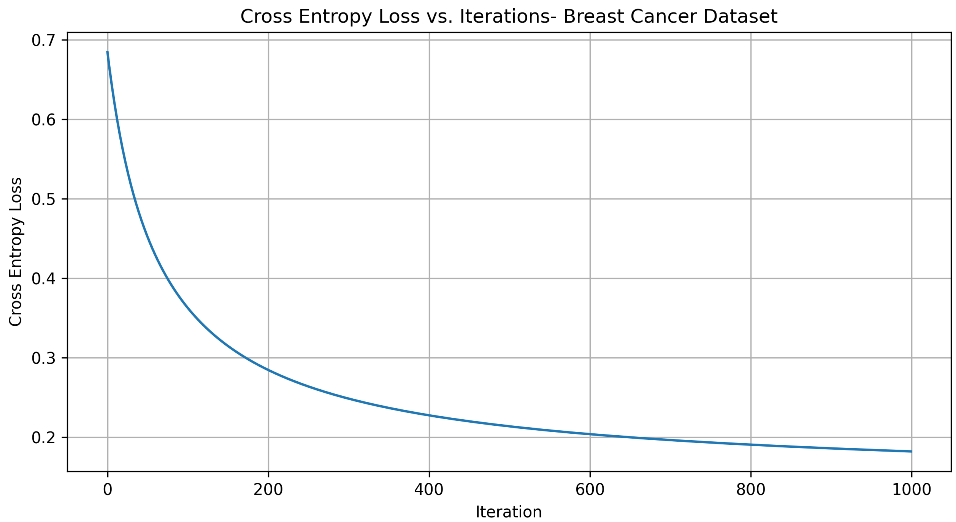
(a) Combined mean absolute value ranking of features for the Wine dataset (Only retained features above the mean threshold).



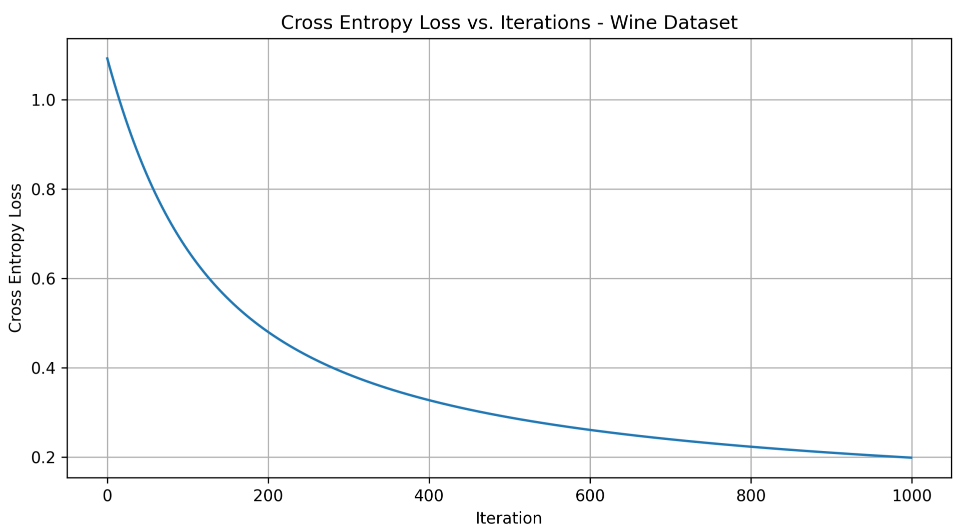
(b) Ranking of features for each class of the Wine dataset.

Figure 2: Feature Importance Rankings for the Wine Dataset.

Next, to verify our logistic regression implementations before training, we compared the analytical gradients to those obtained using numerical perturbation. For the binary logistic regression model, the maximum difference observed was 9×10^{-10} and the mean difference was 4×10^{-10}. For the multi-class logistic regression model, the maximum difference and mean difference were 2.2×10^{-9} and 6×10^{-10} respectively. The maximum difference represents the largest single discrepancy between the two sets of gradients to highlight any potential outlier errors. In contrast, the mean difference is the average of all discrepancies across the gradient components to provide an overall measure of the typical error in our gradient computations. Both of these values are extremely small, which allows us to confirm the correct implementation of our analytical gradients. Additionally, we also plotted cross-entropy loss on the training data as a function of iterations to ensure a smooth decreasing trend (Fig. 3). The convergence plots show a stable decrease in cross-entropy loss over iterations at a learning rate of 0.005 (which was the learning rate shown to achieve the highest test accuracy upon experimentation), indicating the training process was effective.



(a) Cross-Entropy loss as a function of iterations for the Breast Cancer dataset.



(b) Cross-Entropy loss as a function of iterations for the Wine dataset.

Figure 3: Cross-Entropy loss as a function of iterations for the Breast Cancer and Wine datasets.

After confirming that our training process worked correctly, we explored the impact of different train-validation-test splits on model performance. We experimented with four splits: 60/20/20, 65/25/10, 70/15/15, and 80/10/10 to maximize the AUROC for binary classification and validation accuracy for multi-class classification. For the

Breast Cancer dataset (binary classification), our logistic regression model achieved a very high validation AUROC of 0.9979 with the 60/20/20 split. The validation AUROC dropped slightly to 0.9866, 0.9833, and 0.9832 for the 65/25/10, 70/15/15, and 80/10/10 splits, respectively. For the Wine dataset (multi-class classification), both the multi-class logistic regression and the multivariate linear regression models achieved a perfect validation accuracy (100%) with the 60/20/20 and 70/15/15 splits. However, when using the 80/10/10 split, the accuracy of the multi-class logistic model and multivariate linear model dropped to 92.86%. Furthermore, when using the 65/25/10 split, the multi-class logistic model achieved 94.29% validation accuracy and the multivariate linear model achieved 97.14% validation accuracy. These results suggest that split proportions can affect model performance. For example, allocating a larger portion of data to the training set may lead to a smaller validation set that might not fully capture the full dataset's distribution, potentially causing a slight decrease in performance, such as for the 80/10/10 split. Additionally, increasing the validation set too much at the expense of the training set could hinder the model's ability to learn effectively. Overall, this experiment allowed us to determine that the 60/20/20 split provided the best validation performance across all models, so we used it for our final test evaluations. We found that for the binary classification (Breast Cancer dataset), multiple linear regression and binary logistic regression both performed well with test AUROCs of 0.9987 and 0.9974 respectively. For multi-class classification (Wine dataset), multivariate and multiple linear regression achieved test accuracies of 100% and multi-class logistic regression reached a test accuracy of 94.44%.

To directly compare the binary models, we plotted their ROC curves (Fig. 4), which showed that both models classified extremely well, with the linear regression model's AUROC being only 0.0013 higher than that of the logistic regression model.

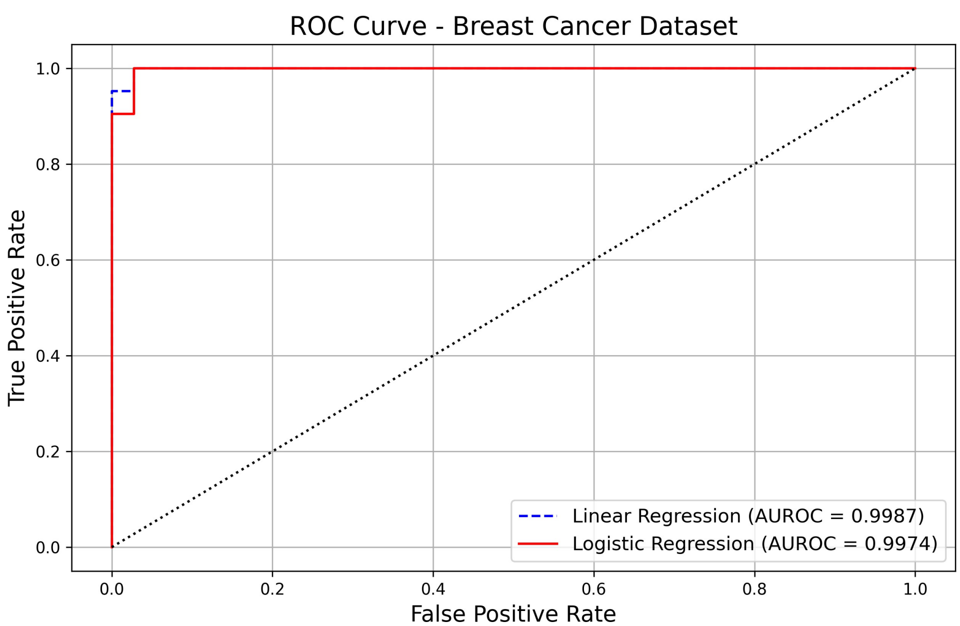


Figure 4: Comparison of ROC and AUROC scores for binary logistic regression and linear regression models.

We also compared the consistency of feature importance between modeling approaches by comparing the binary regression coefficients obtained from our simple linear regression method with those learned by our binary logistic regression model. The linear coefficients were computed using the simple regression formula $w_d = \frac{x_d^\top y}{N}$, while the logistic regression coefficients were derived from our cross-entropy minimization procedure. The horizontal bar plots (Fig. 5) display the coefficients side by side. The linear regression coefficients range from 0.567 to 0.799, whereas the logistic regression coefficients are consistently lower, ranging from about 0.195 to 0.385. While the absolute magnitudes differ—which is expected given the different loss functions and scaling factors—the ranking of features remains roughly consistent between the two methods. This indicates that both models identify similar features as being the most influential for the binary classification task. The observed differences in the range of values likely arise from the differences in the modeling approaches (least squares vs. cross-entropy). Overall, these findings reinforce our feature selection process and suggest that the associations between the features and the target variable are reliable across both linear and logistic regression models.

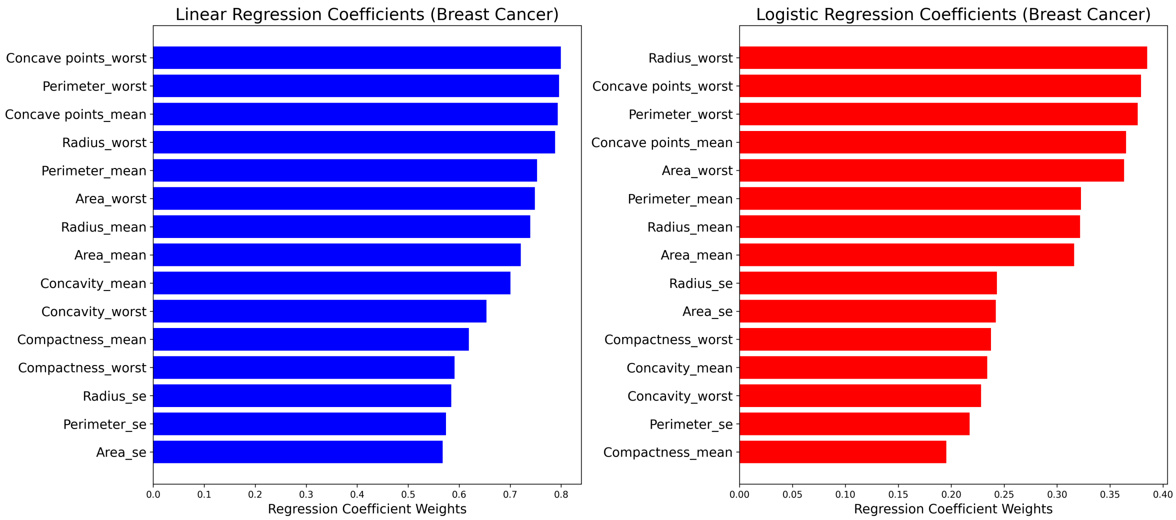


Figure 5: Comparison of simple linear regression coefficients and binary logistic regression coefficients for the Breast Cancer dataset.

Similarly, we also compared the $D \times C$ coefficient matrices obtained from our multi-class logistic regression model and our simple regression-based (multiple linear regression) approach for the Wine Recognition dataset. To aid with this comparison, we plotted the two heatmaps side by side (Fig. 6). The first heatmap visualizes the coefficients computed using the function $W = \frac{X^\top Y}{N}$, while the other displays the coefficients learned by our multi-class logistic regression model. In both heatmaps, the rows represent the features and the columns represent the classes. The color scale, centered at zero, allows us to see both positive and negative associations (red is positive, blue is negative). Our results indicate that the overall ranking of features is quite similar, with small differences in the coefficient values. These differences are likely due to the distinct loss functions and optimization processes used by the two methods.

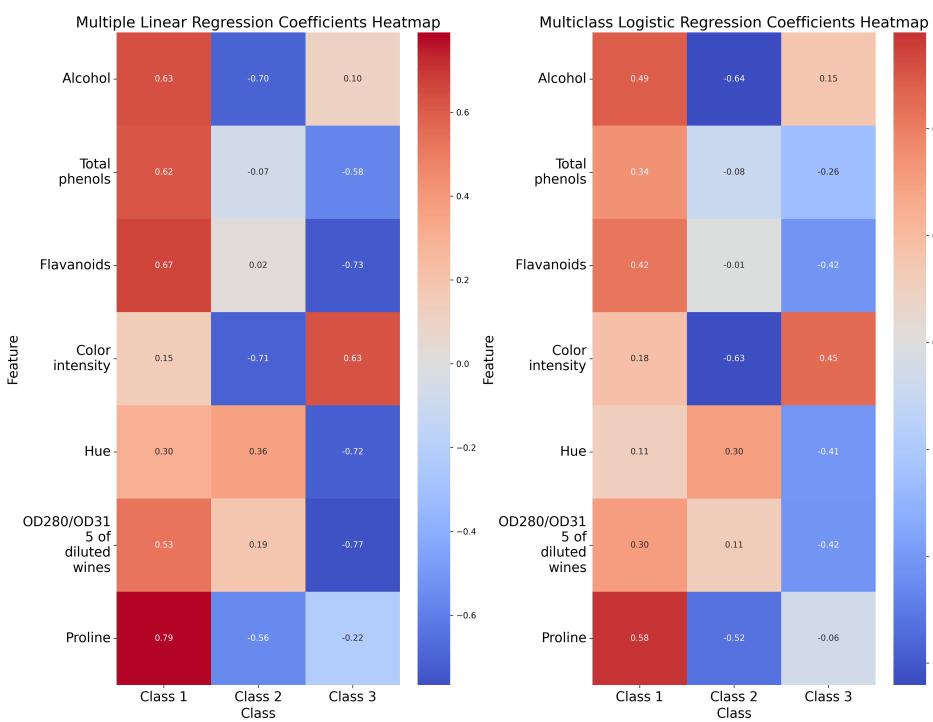


Figure 6: Heatmap comparison of (simple) multiple linear regression coefficients and multi-class logistic regression coefficients for the Wine dataset.

To try and improve our logistic regression models, we used a non-linear feature transformation using a degree-2 polynomial basis. This transformation expands the original feature set into a higher-dimensional space to hopefully allow our logistic regression models to capture interactions and non-linear effects that may not be clear in the original features. Before transformation, our binary logistic regression achieved a test AUROC of 0.9974, and after transformation the AUROC was 0.9987, which is equivalent to the AUROC of the linear regression model. For the multi-class logistic model, we achieved a test accuracy of 94.44% before transformation and a test accuracy of 100% after. This demonstrates the value of feature transformation in enhancing model flexibility and performance. Lastly, we compared our regression models against standard classification models such as K-Nearest Neighbors (KNN) and Decision Trees (DT). We implemented KNN and DT from scikit-learn. Using the Breast Cancer dataset, KNN achieved an AUROC of 0.9974, which matches the AUROC of our logistic regression model without transforming features. Decision Tree reached a slightly lower AUROC of 0.9385. However, for the Wine dataset, KNN achieved an accuracy of 94.44% while DT performed better with an accuracy of 100%. These results are very similar to the results achieved using our custom regression models, providing further confidence in the implementations of our models.

5 Discussion and Conclusion

These experiments highlighted several findings about our regression models and the classification methods in general. Overall, our multiple linear regression model outperformed our binary logistic regression model slightly for the Breast Cancer dataset. Similarly, our multivariate and multiple linear regression models achieved a higher test accuracy for the Wine dataset than our multi-class logistic regression model. However, applying a non-linear feature transformation with second-degree polynomial for our logistic regression models slightly improved their performance. While both linear and logistic models performed very well, one reason that our linear regression models could have outperformed our logistic regression models is that there is an underlying linear relationship between the features and the target, making the least squares loss function very effective. Although cross-entropy loss is generally more appropriate for classification because it is designed to optimize probability estimates, the use of a sigmoid transformation to obtain probabilities for our linear models seems to have captured the decision boundary slightly more accurately.

When comparing the regression coefficients, both models tend to rank the most important features similarly. For example, Concave points_worst, Perimeter worst_and Concave points_mean consistently appear at the top. However, the magnitudes differ with logistic regression coefficients being smaller due to the scaling imposed by the cross-entropy loss. While least squares minimizes the squared differences directly, cross-entropy adjusts the coefficients to best match the predicted probabilities to the true class distribution.

The top features also make sense in real world applications. For example, for the Breast Cancer dataset, concave points are indentations in the border of a tumor that can indicate its malignancy. Tumors with more concave points are more likely to be cancerous [3], which explains why it is one of the top informative features for whether a tumor is benign or malignant. Furthermore, proline, color intensity, and alcohol are well recognized features for assessing wine quality [1]. Overall, our results show that carefully selecting and transforming features can lead to highly accurate classification performance using both linear and logistic regression models.

6 Statement of Contributions

All group members produced unique code and the final Google Colab is a cumulation of everyone's code. Genevieve drafted the report with editing done by Luke and Luca.

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